

Test 31: Hatano-Nelson non-Hermitian skin

Verdict: PASS | Generated: 2026-09-15 05:29:51 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Deterministic non-Hermitian spectral property: complex eigenvalue count/pattern.

METHOD: Deterministic non-Hermitian spectral property: complex eigenvalue count/pattern.

Test 31 SERIES: n=5 realizations $\times \{0.5, 0.6, 0.7, 0.8\}$ @48x48, n_cx mean 0,2225,2281,2295, $r \in [0.8953, 0.9253]$, max|Im| 0.3184-0.9576.

Result [SERIES pass 3]: 4 sub-checks, 0 failed ? PASS

- Plots
- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_04.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 31 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.
Independent verification guide

[05:25:20.846] [+20311.875s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[05:25:24.130] [+20315.159s] calc: $r_{NN} = 0.894102$ (n=2304, Poisson-box ref NaN, 10 clouds)

[05:25:30.843] [+20319.122s] calc: $r_{NN} = 0.907204$ (n=2304, Poisson-box ref 0.851687, 10 clouds)

[05:25:56.656] [+20347.685s] calc: $r_{NN} = 0.925299$ (n=2304, Poisson-box ref 0.852026, 10 clouds)

[05:26:13.442] [+20364.470s] calc: $r_{NN} = 0.930754$ (n=2304, Poisson-box ref 0.850043, 10 clouds)

[05:26:14.212] [+20365.240s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[05:26:17.814] [+20368.842s] calc: $r_{NN} = 0.887760$ (n=2304, Poisson-box ref NaN, 10 clouds)

[05:26:33.688] [+20384.716s] calc: $r_{NN} = 0.910544$ (n=2304, Poisson-box ref 0.852173, 10 clouds)

[05:26:50.420] [+20401.449s] calc: $r_{NN} = 0.923175$ (n=2304, Poisson-box ref 0.852783, 10 clouds)

[05:27:06.833] [+20417.861s] calc: $r_{NN} = 0.930952$ (n=2304, Poisson-box ref 0.850774, 10 clouds)

[05:27:07.596] [+20418.624s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[05:27:11.143] [+20422.171s] calc: $r_{NN} = 0.898821$ (n=2304, Poisson-box ref NaN, 10 clouds)

[05:27:26.749] [+20437.777s] calc: $r_{NN} = 0.878783$ (n=2304, Poisson-box ref 0.851955, 10 clouds)

[05:27:43.994] [+20455.023s] calc: $r_{NN} = 0.888162$ (n=2304, Poisson-box ref 0.852427, 10 clouds)

[05:28:01.042] [+20472.071s] calc: $r_{NN} = 0.891964$ (n=2304, Poisson-box ref 0.850764, 10 clouds)

[05:28:01.794] [+20472.823s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[05:28:05.354] [+20476.382s] calc: ?r?_NN = 0.897703 (n=2304, Poisson-box ref NaN, 10 clouds)

[05:28:21.797] [+20492.826s] calc: ?r?_NN = 0.903503 (n=2304, Poisson-box ref 0.851913, 10 clouds)

[05:28:38.374] [+20509.402s] calc: ?r?_NN = 0.923336 (n=2304, Poisson-box ref 0.852584, 10 clouds)

[05:28:54.601] [+20525.629s] calc: ?r?_NN = 0.935506 (n=2304, Poisson-box ref 0.850682, 10 clouds)

[05:28:55.388] [+20526.416s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[05:28:58.950] [+20529.978s] calc: ?r?_NN = 0.898157 (n=2304, Poisson-box ref NaN, 10 clouds)

[05:29:15.117] [+20546.146s] calc: ?r?_NN = 0.906060 (n=2304, Poisson-box ref 0.851688, 10 clouds)

[05:29:32.225] [+20563.253s] calc: ?r?_NN = 0.920104 (n=2304, Poisson-box ref 0.851832, 10 clouds)

[05:29:50.149] [+20581.177s] calc: ?r?_NN = 0.937177 (n=2304, Poisson-box ref 0.850592, 10 clouds)

CONSOLE CAPTURE

??

TEST 31 (pass 3, SERIES): Hatano?Nelson ? realization ensemble

??

Ensemble: n=5 realizations x? ? {0.5, 0.6, 0.7, 0.8}, ? = 5.0

Lattice: 48x48 (N=2304), W_eff = 1.00 (standard), Nv=4 neutral, q=-1.00

? RUNTIME: 15 zgeev + 5 heevr rows at 48? ? 1?2 min

real 1/5 ?=0.5: n_complex = 0, max|Im| = 0.0000, ?r?NN = 0.8941 (2.9 s)

real 1/5 ?=0.6: n_complex = 2243, max|Im| = 0.3195, ?r?NN = 0.9072 (15.5 s)

real 1/5 ?=0.7: n_complex = 2283, max|Im| = 0.6397, ?r?NN = 0.9253 (16.8 s)

real 1/5 ?=0.8: n_complex = 2298, max|Im| = 0.9602, ?r?NN = 0.9308 (16.8 s)

real 2/5 ?=0.5: n_complex = 0, max|Im| = 0.0000, ?r?NN = 0.8878 (3.2 s)

real 2/5 ?=0.6: n_complex = 2237, max|Im| = 0.3194, ?r?NN = 0.9105 (15.9 s)

real 2/5 ?=0.7: n_complex = 2297, max|Im| = 0.6396, ?r?NN = 0.9232 (16.7 s)

real 2/5 ?=0.8: n_complex = 2299, max|Im| = 0.9598, ?r?NN = 0.9310 (16.4 s)

real 3/5 ?=0.5: n_complex = 0, max|Im| = 0.0000, ?r?NN = 0.8988 (3.1 s)

real 3/5 ?=0.6: n_complex = 2147, max|Im| = 0.3147, ?r?NN = 0.8788 (15.6 s)

real 3/5 ?=0.7: n_complex = 2239, max|Im| = 0.6319, ?r?NN = 0.8882 (17.2 s)

real 3/5 ?=0.8: n_complex = 2274, max|Im| = 0.9501, ?r?NN = 0.8920 (17.0 s)

real 4/5 ?=0.5: n_complex = 0, max|Im| = 0.0000, ?r?NN = 0.8977 (3.1 s)

real 4/5 ?=0.6: n_complex = 2260, max|Im| = 0.3199, ?r?NN = 0.9035 (16.4 s)

real 4/5 ?=0.7: n_complex = 2300, max|Im| = 0.6399, ?r?NN = 0.9233 (16.6 s)

real 4/5 ?=0.8: n_complex = 2303, max|Im| = 0.9601, ?r?NN = 0.9355 (16.2 s)

real 5/5 ?=0.5: n_complex = 0, max|Im| = 0.0000, ?r?NN = 0.8982 (3.1 s)

real 5/5 ?=0.6: n_complex = 2237, max|Im| = 0.3183, ?r?NN = 0.9061 (16.2 s)

real 5/5 ?=0.7: n_complex = 2287, max|Im| = 0.6381, ?r?NN = 0.9201 (17.1 s)

real 5/5 ?=0.8: n_complex = 2303, max|Im| = 0.9578, ?r?NN = 0.9372 (17.9 s)

Ensemble means: ?=0.5: n_cx=0.0, max|Im|=0.00e+00, ?r?=0.8953

?=0.8: n_cx=2295.4, max|Im|=0.9576, ?r?=0.9253 | Poisson-box baseline ? 0.8500

S1 anchor: OK ? | S2 sweep: OK ? | S3 ?r? rise: OK ? (0.8953?0.9253) | S4 max|Im| grow: OK ? (0.3184?0.9576)

??

?=0.5 Hermitian ?real PASS ? n_complex = 0 and max|Im| < 1e-10 in all 5 realizations

n_complex > 0 ?(? ,k) PASS ? every non-Hermitian row (5 ×3) develops complex eigenvalues

?r? rises with ? (2-D repulsion) PASS ? ensemble mean 0.8953 (?=0.5, line) ? 0.9253 (?=0.8, area)

max|Im| grows with ? PASS ? ensemble mean 0.3184 (?=0.6) ? 0.9576 (?=0.8), g = (?%?)•ln|?

Test 31c [SERIES pass 3]: 4 sub-checks, 0 failed ? PASS